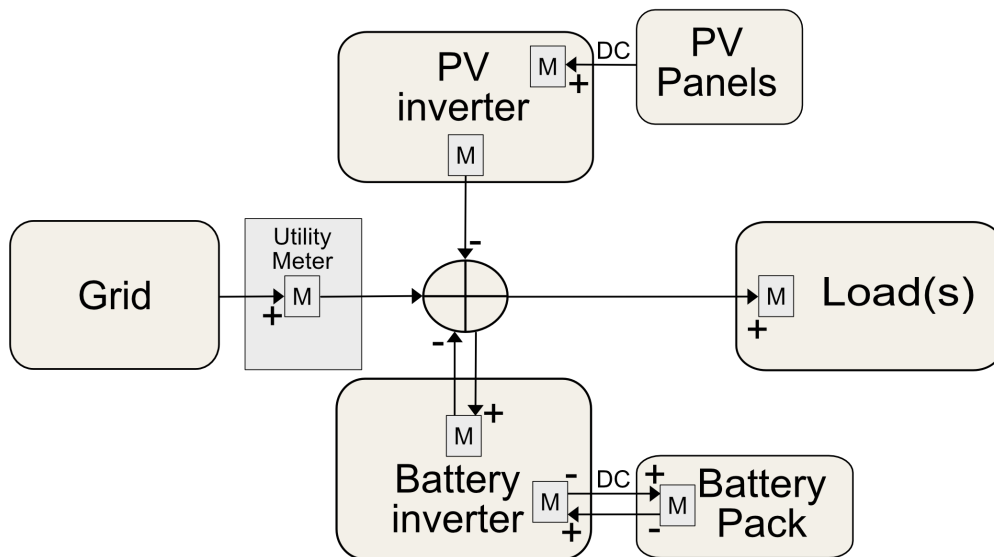


9.2.6. Definitions

9.2.6.1. Power

Power is defined in the main specification (see Data Model) in units of milliwatts. It is a signed value, where positive values indicate the direction of power flow into the node. A negative value indicates that the device is supplying power, and thus reducing the overall load of the premises on the grid supply.



*NOTE: Charging the battery implies a positive power
Discharging the battery implies a negative power
from the perspective of the meter 'M' inside the node*

Figure 21. Power flows into devices

Solar PV inverter example: A solar PV inverter is normally connected to the premises' wiring, so its generation power will be indicated as a negative integer, since power is flowing out of the node towards the mains connection. Note that at night (when there is no solar production) the PV inverter will consume some standby power, and this will result in a small positive power reading.

Grid Power: Power from the grid is measured by a utility meter which is imported into the node, so positive power values indicate that power is flowing into the premises. Negative values indicate power is flowing back towards the grid.

EVSE example: An EVSE provides power to the EV (when charging) so a positive value indicates that the power is flowing into the EVSE from the mains supply when the EV is charging, and is acting as a load on the grid supply, and negative value indicates that the EV is discharging.

BESS example: A battery storage inverter normally provides power to loads when discharging, so negative power indicates discharging (power is flowing back to the mains supply) therefore reducing the load on the grid, and positive power indicates power flowing from the mains to the battery during charging (adding load to the grid).

Washing Machine example: A washing machine only consumes power (i.e. is a load), so it will always have a positive power value.